STA106_Midterm2_Project_RCode

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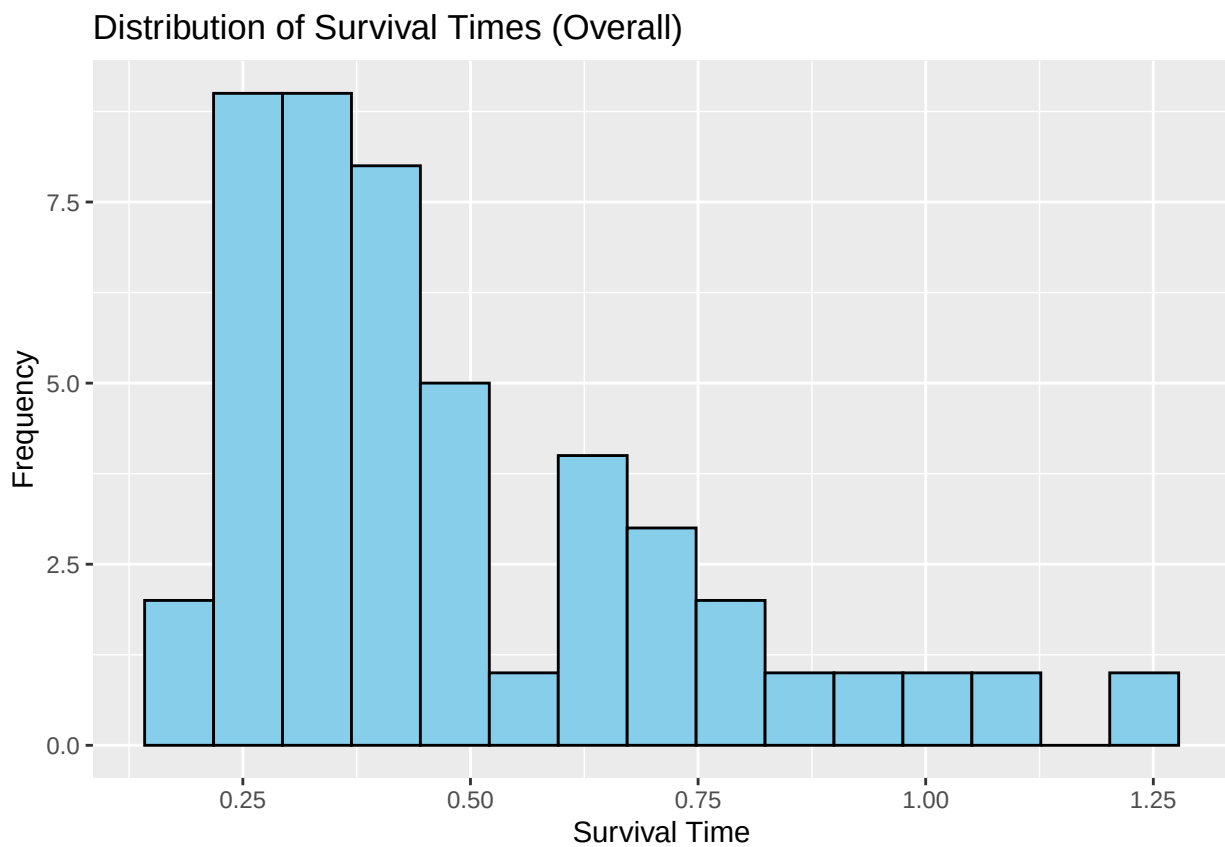
Load the data

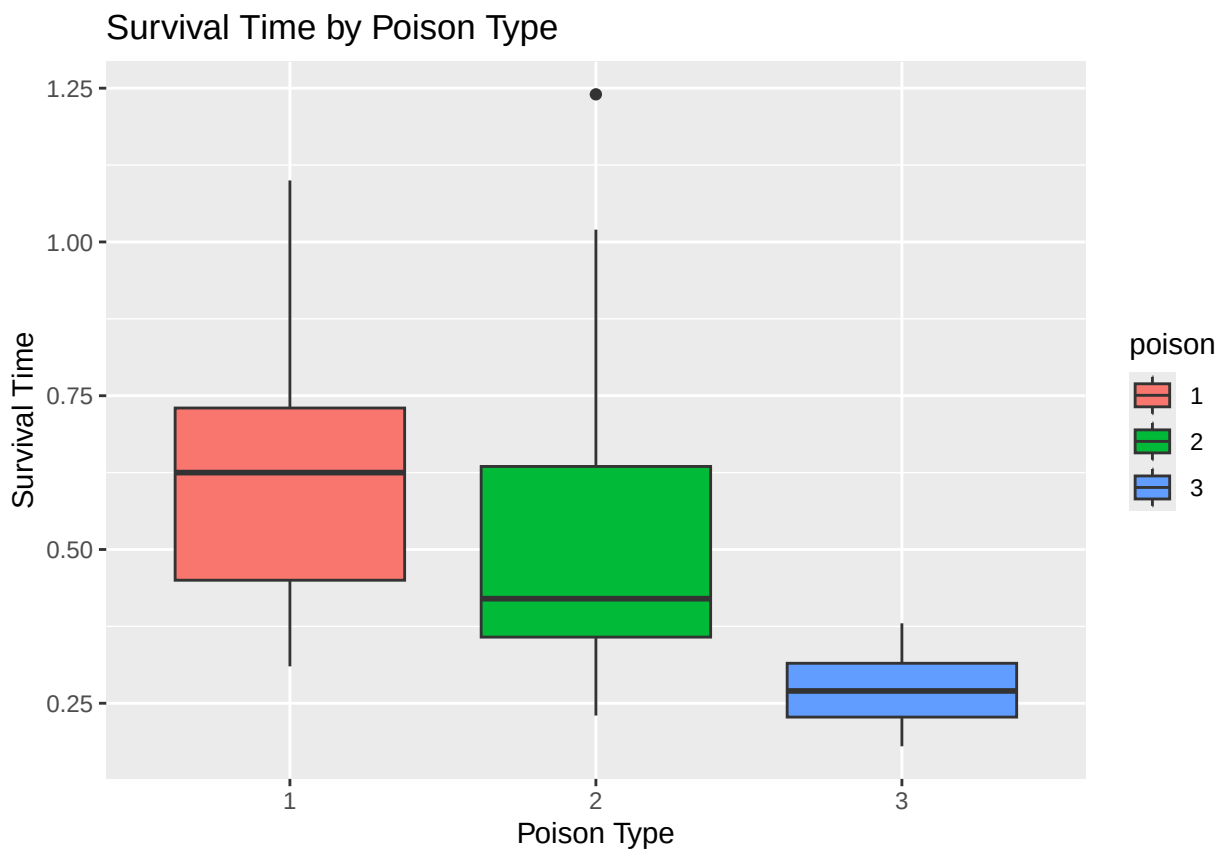
```r
df <- read.csv("poisons.csv")

# poison and treatment are factors
df$poison <- factor(df$poison)
df$treat <- factor(df$treat)
```

Figures

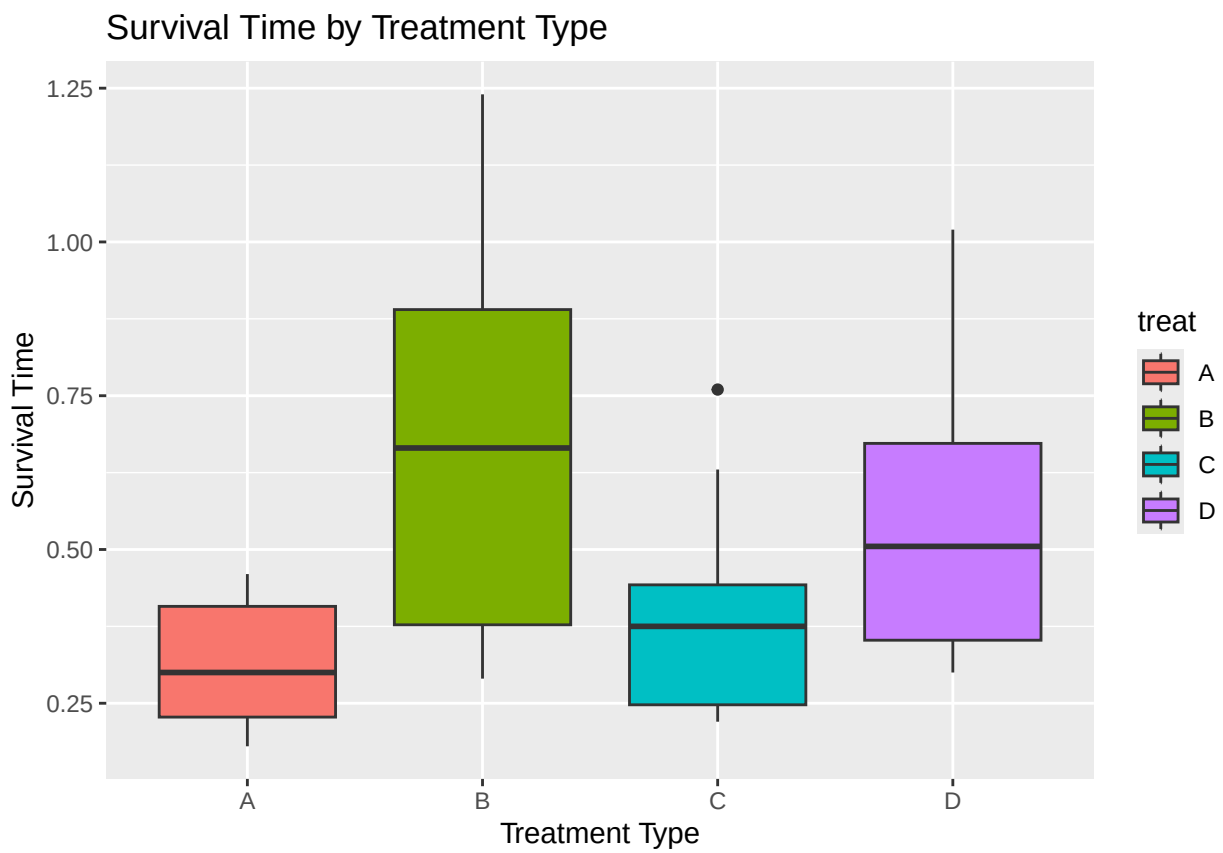
```r
# Figure 1: Overall Histogram of Survival Time
 ggplot(df, aes(x = time)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 15) +
  labs(title = "Distribution of Survival Times (Overall)",
       x = "Survival Time",
       y = "Frequency")
```



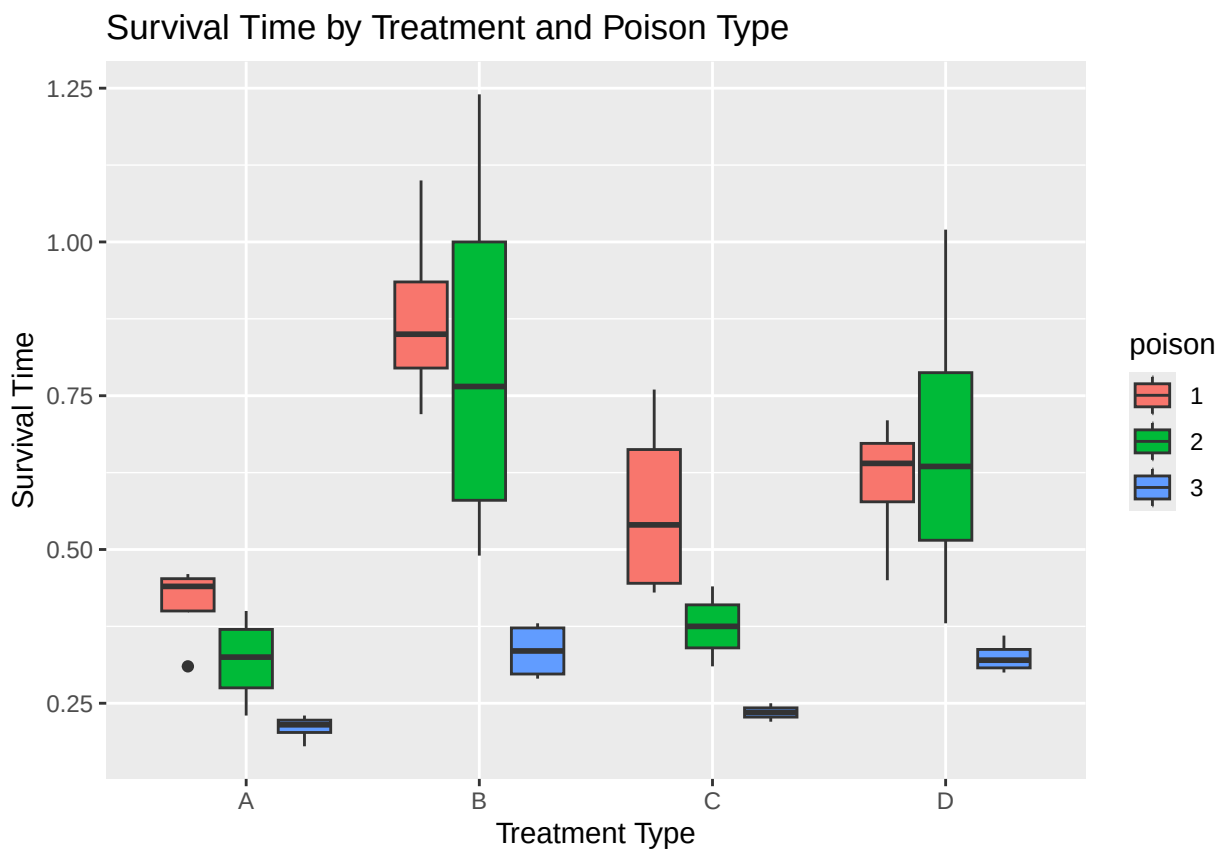

```r
# Figure 2: Boxplot by Poison Type
ggplot(df, aes(x = poison, y = time, fill = poison)) +
  geom_boxplot() +
  labs(title = "Survival Time by Poison Type",
       x = "Poison Type",
       y = "Survival Time")
```




```r
# Figure 3: Boxplot by Treatment Type
ggplot(df, aes(x = treat, y = time, fill = treat)) +
  geom_boxplot() +
  labs(title = "Survival Time by Treatment Type",
       x = "Treatment Type",
       y = "Survival Time")
```




```r
# Figure 4:survival Time by Treatment and Poison Type
ggplot(df, aes(x = treat, y = time, fill = poison)) +
  geom_boxplot() +
  labs(title = "Survival Time by Treatment and Poison Type",
       x = "Treatment Type",
       y = "Survival Time")
```


Survival Time by Treatment and Poison Type
1.25
1.00
0.75
0.50
0.25
Survival Time
A
B
C
D
Treatment Type
poison
1
2
3

Summaries

```r
# 1. Grouped Summary (Poison & Treatment)
summary_stats <- df %>%
  group_by(poison, treat) %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

# 2. Summary by Poison Type
poison_summary <- df %>%
  group_by(poison) %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

# 3. Summary by Treatment Type
treat_summary <- df %>%
  group_by(treat) %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

# 4. Overall Summary
overall_stats <- df %>%
  summarise(
    Mean_Time = mean(time, na.rm = TRUE),
    SD_Time = sd(time, na.rm = TRUE),
    n = n()
  )
```

Tables

```r
# Table 1: Overall
knitr::kable(
  overall_stats,
  booktabs = TRUE,
  digits = 4,
  caption = "Overall Summary Statistics",
  col.names = c("Overall Mean Time", "Overall Std. Deviation", "Total Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 1: Overall Summary Statistics

Overall Mean Time	Overall Std. Deviation	Total Sample Size
0.4794	0.2529	48

```r
# Table 2: Grouped
knitr::kable(
  summary_stats,
  booktabs = TRUE,
  digits = 4,
  caption = "Summary Statistics by Poison and Treatment",
  col.names = c("Poison", "Treatment", "Mean Time", "Std. Deviation", "Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 2: Summary Statistics by Poison and Treatment

Poison	Treatment	Mean Time	Std. Deviation	Sample Size
1	A	0.4125	0.0695	4
1	B	0.8800	0.1608	4
1	C	0.5675	0.1567	4
1	D	0.6100	0.1128	4
2	A	0.3200	0.0753	4
2	B	0.8150	0.3363	4
2	C	0.3750	0.0569	4
2	D	0.6675	0.2710	4
3	A	0.2100	0.0216	4
3	B	0.3350	0.0465	4
3	C	0.2350	0.0129	4
3	D	0.3250	0.0265	4

```r
# Table 3: By Poison Type
knitr::kable(
  poison_summary,
  booktabs = TRUE,
  digits = 4,
```

```r
  caption = "Summary Statistics by Poison Type",
  col.names = c("Poison Type", "Mean Time", "Std. Deviation", "Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 3: Summary Statistics by Poison Type

Poison Type	Mean Time	Std. Deviation	Sample Size
1	0.6175	0.2094	16
2	0.5444	0.2894	16
3	0.2762	0.0623	16

```r
# Table 4: By Treatment Type
knitr::kable(
  treat_summary,
  booktabs = TRUE,
  digits = 4,
  caption = "Summary Statistics by Treatment Type",
  col.names = c("Treatment Type", "Mean Time", "Std. Deviation", "Sample Size")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 4: Summary Statistics by Treatment Type

Treatment Type	Mean Time	Std. Deviation	Sample Size
A	0.3142	0.1023	12
B	0.6767	0.3208	12
C	0.3925	0.1670	12
D	0.5342	0.2194	12

Diagnosistic

```r
# fit initial model
model_initial <- lm(time ~ poison * treat, data = df)

# 1. Outlier Check with Studentized Residuals and Bonferroni cutoff
nt <- nrow(df)
a <- length(unique(df$poison))
b <- length(unique(df$treat))
df_resid <- nt - (a * b)
alpha <- 0.05
t.cutoff <- qt(1 - alpha / (2 * nt), df_resid)
cat("cutoff value = ", t.cutoff, "\n")
```

```
## cutoff value =  3.567639
```

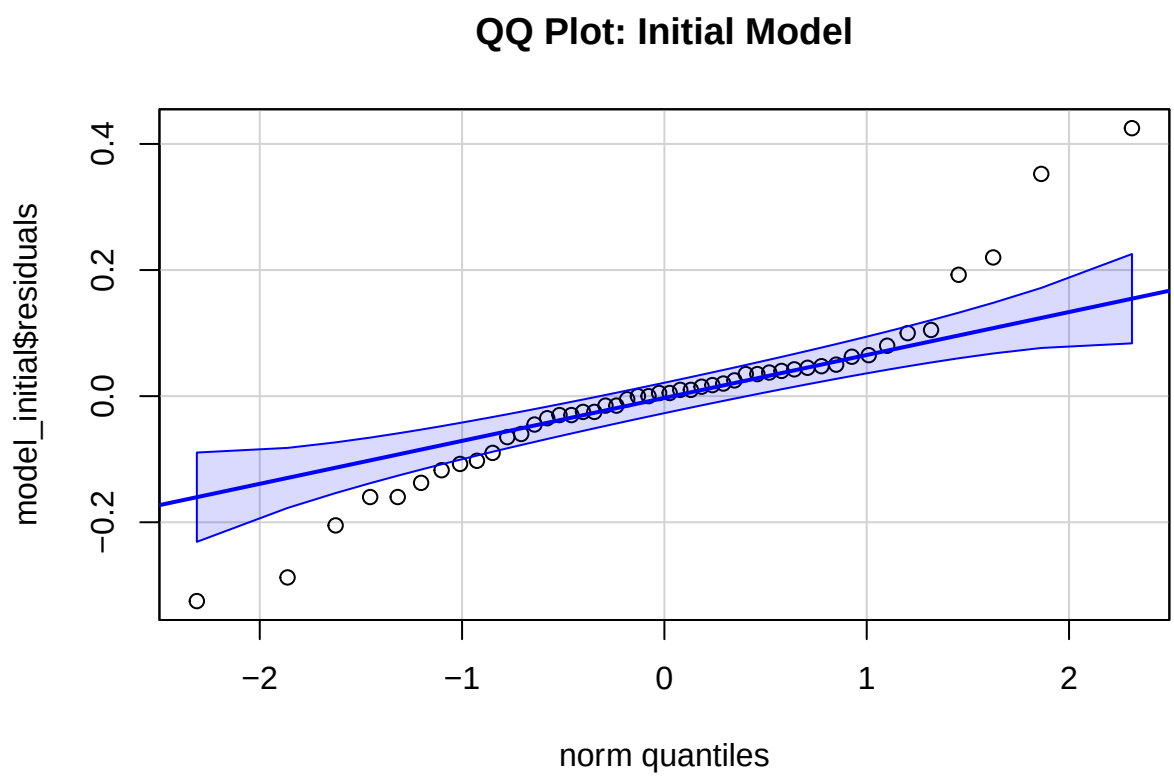
```r
# Outlier via semi-studentized/standardized residuals
MSE <- sum(model_initial$residuals^2) / df_resid
semi_studentized <- model_initial$residuals / sqrt(MSE)
outliers_semi <- which(abs(semi_studentized) > t.cutoff)
cat("Outliers via Semi-studentized residuals:", paste(outliers_semi, collapse=", "), "\n")
```

```
## Outliers via Semi-studentized residuals:
```

```r
# Outliers via studentized/standardized residuals (rstandard)
rij <- rstandard(model_initial)
outliers_std <- which(abs(rij) > t.cutoff)
cat("Outliers via studentized residuals (rstandard):", paste(outliers_std, collapse=", "), "\n\n")
```

```
## Outliers via studentized residuals (rstandard):
```

```r
# 2. Normality Check
car::qqPlot(model_initial$residuals, id = FALSE, main = "QQ Plot: Initial Model")
```

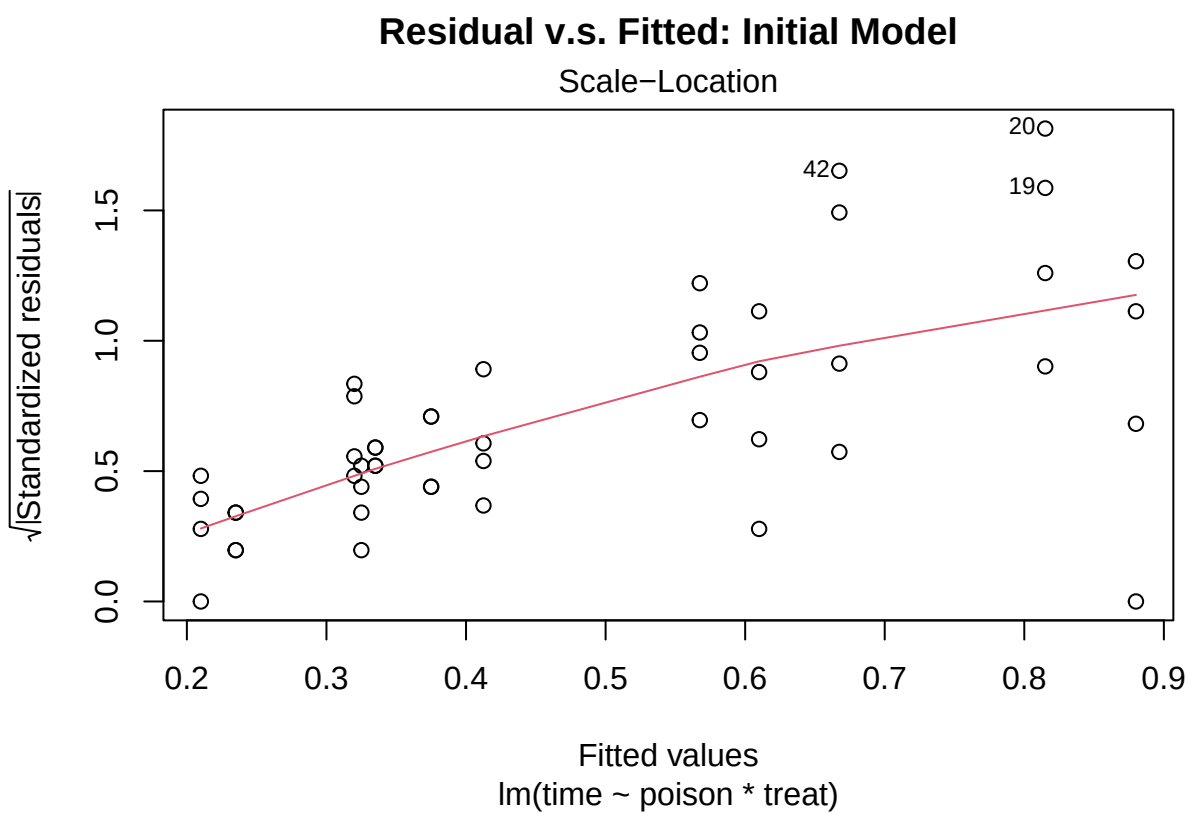



```r
shapiro_initial <- shapiro.test(model_initial$residuals)
print(shapiro_initial)
```

```
##
## Results of Hypothesis Test
```

```
## --------------------------
##
## Alternative Hypothesis:
##
## Test Name:                        Shapiro-Wilk normality test
##
## Data:                             model_initial$residuals
##
## Test Statistic:                   W = 0.9122892
##
## P-value:                          0.001606758
```

```r
# 3. Constant Variance Check
plot(model_initial, which = 3, main = "Residual v.s. Fitted: Initial Model")
```

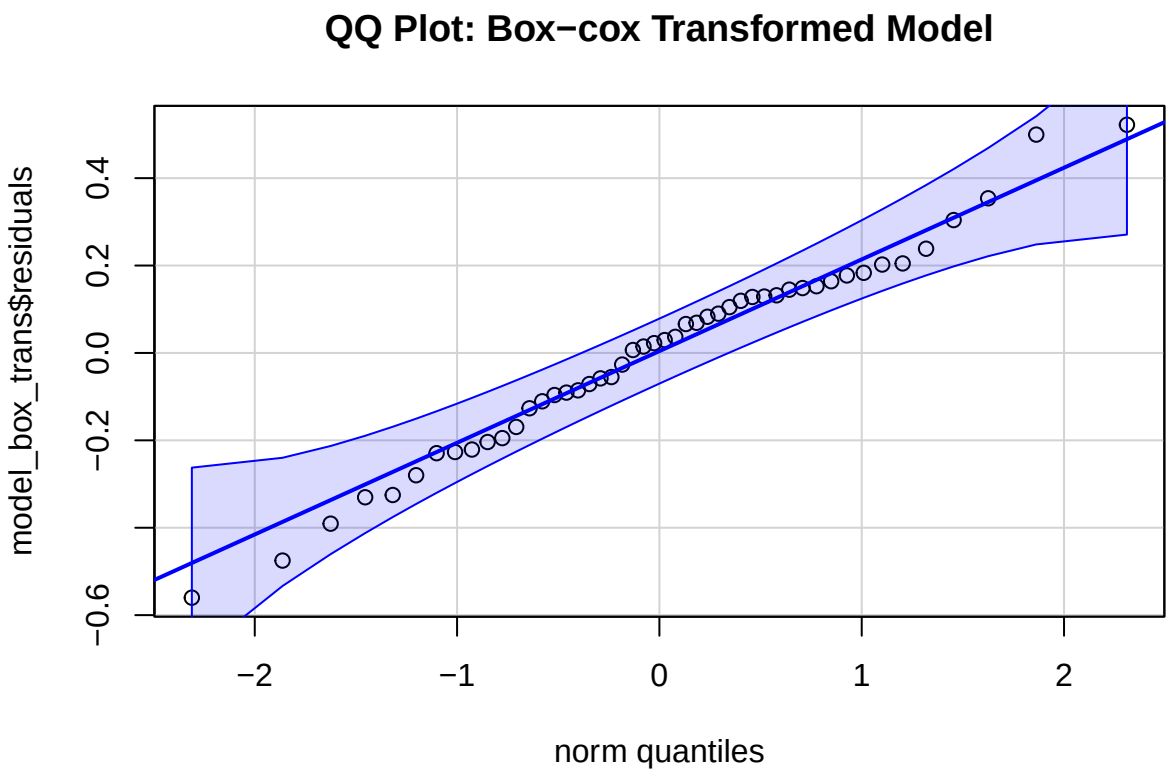



```r
levene_initial <- leveneTest(model_initial)
print(levene_initial)
```

```
##       Df F value       Pr(>F)
## group 11 4.13233 0.0005833475
##       36      NA           NA
```

```r
# 4. Evalutate Two Transformations
# Test Box-Cox Transformation
```

```r
# get best lambda using Shapiro-Wilk
L2 <- boxcox(model_initial, objective.name = "Shapiro-Wilk", optimize = TRUE)$lambda

# apply transformation
df$time_box_trans <- (df$time^L2 - 1) / L2

# fit new model with transformed data
model_box_trans <- lm(time_box_trans ~ poison * treat, data = df)

# redo diagnostics on transformed model
car::qqPlot(model_box_trans$residuals, id = FALSE, main = "QQ Plot: Box-cox Transformed Model")
```

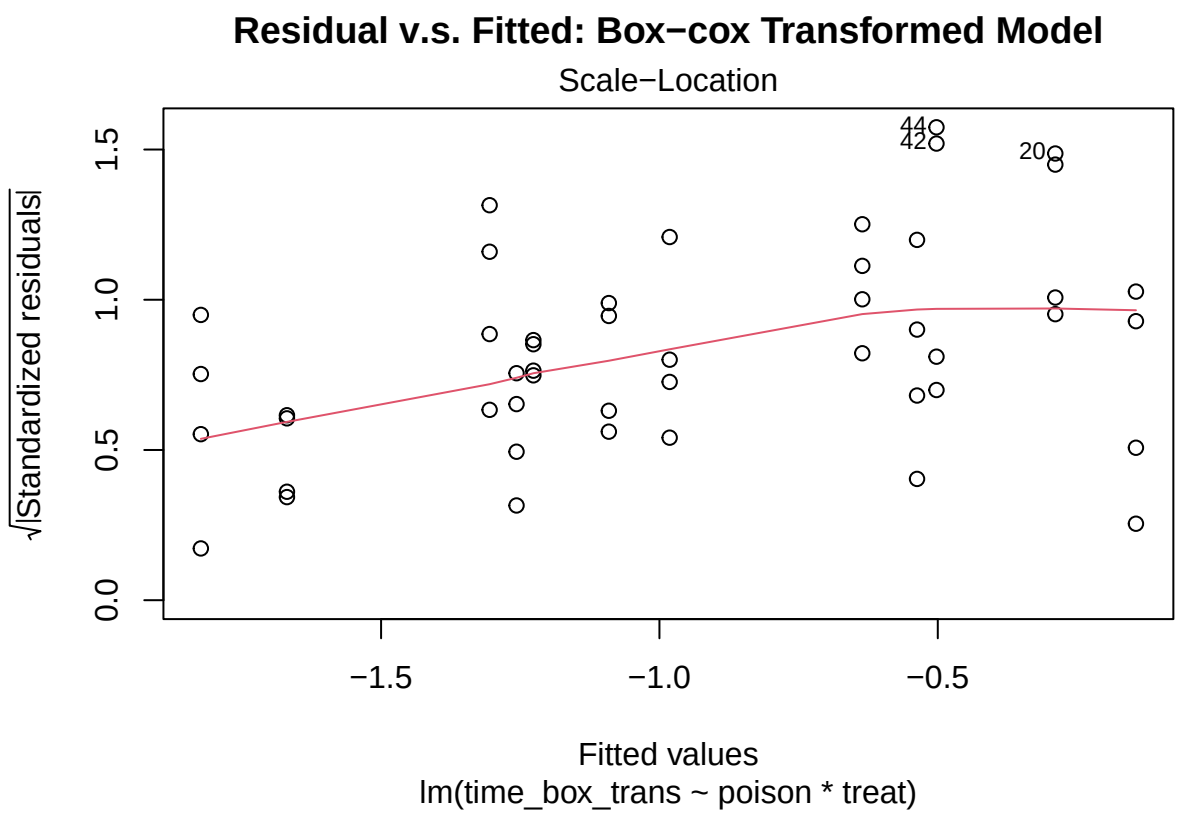



```r
print(shapiro.test(model_box_trans$residuals))
```

```
##
## Results of Hypothesis Test
## --------------------------
##
## Alternative Hypothesis:
##
## Test Name:                      Shapiro-Wilk normality test
##
## Data:                           model_box_trans$residuals
##
## Test Statistic:                 W = 0.9839998
```

```
##
## P-value:                          0.7496128
```

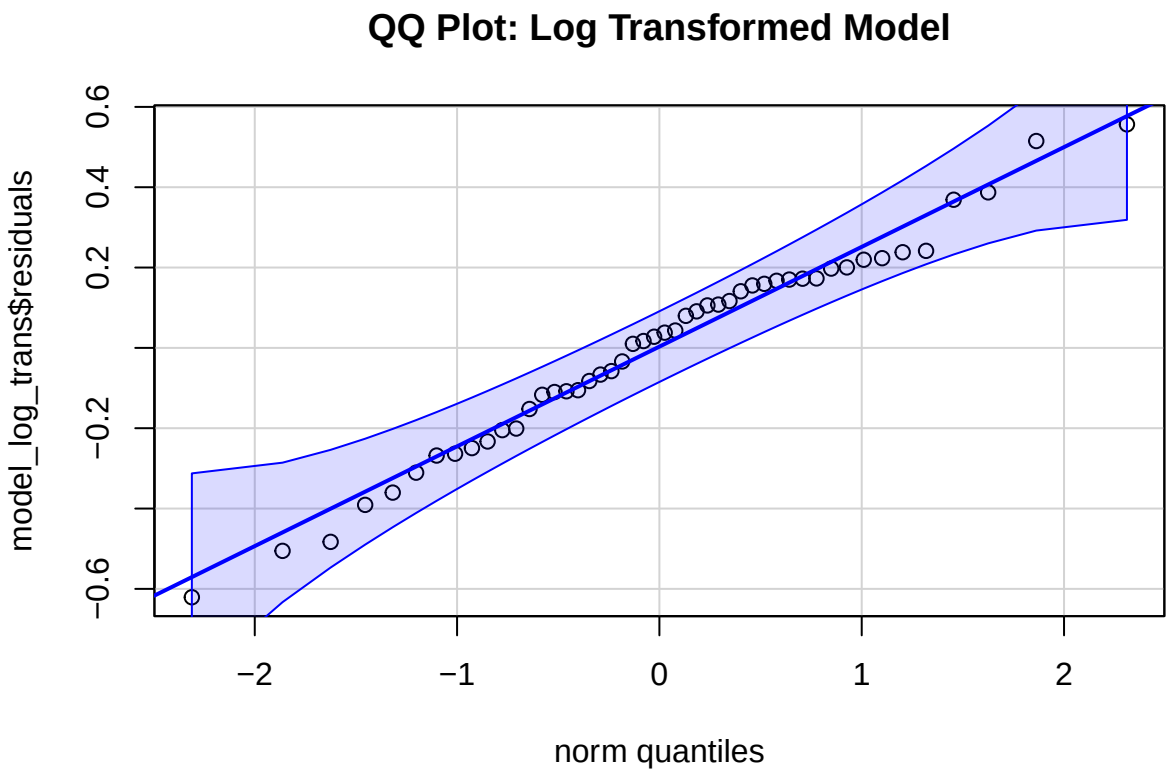
```r
plot(model_box_trans, which = 3, main = "Residual v.s. Fitted: Box-cox Transformed Model")
```




```r
print(leveneTest(model_box_trans))
```

```
##       Df  F value     Pr(>F)
## group 11 2.155772 0.04091845
##       36       NA         NA
```

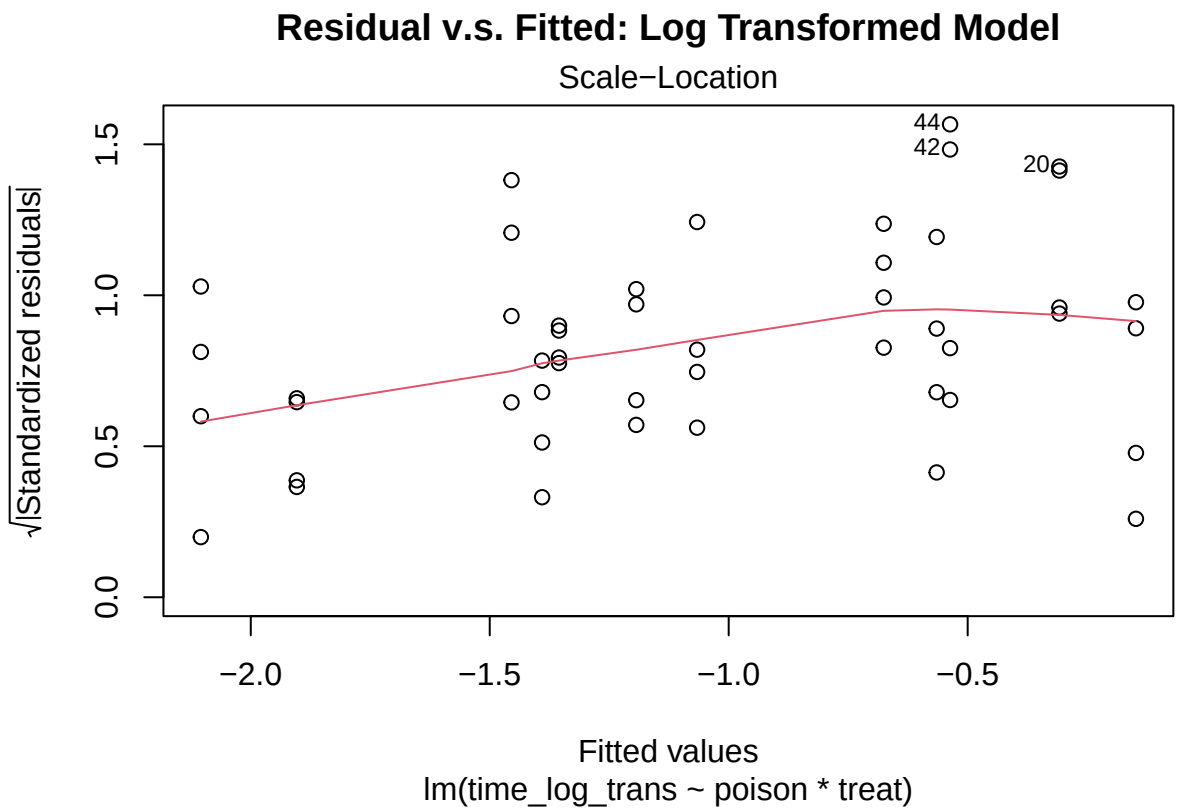
```r
# Test Log Transformation
L3 <- boxcox(df$time, objective.name = "Log-Likelihood", optimize = TRUE)$lambda
df$time_log_trans <- (df$time^L3 - 1) / L3
model_log_trans <- lm(time_log_trans ~ poison * treat, data = df)

car::qqPlot(model_log_trans$residuals, id = FALSE, main = "QQ Plot: Log Transformed Model")
```




```r
print(shapiro.test(model_log_trans$residuals))
```

```
##
## Results of Hypothesis Test
## --------------------------
##
## Alternative Hypothesis:
##
## Test Name:                      Shapiro-Wilk normality test
##
## Data:                           model_log_trans$residuals
##
## Test Statistic:                 W = 0.9820846
##
## P-value:                        0.6673954
```

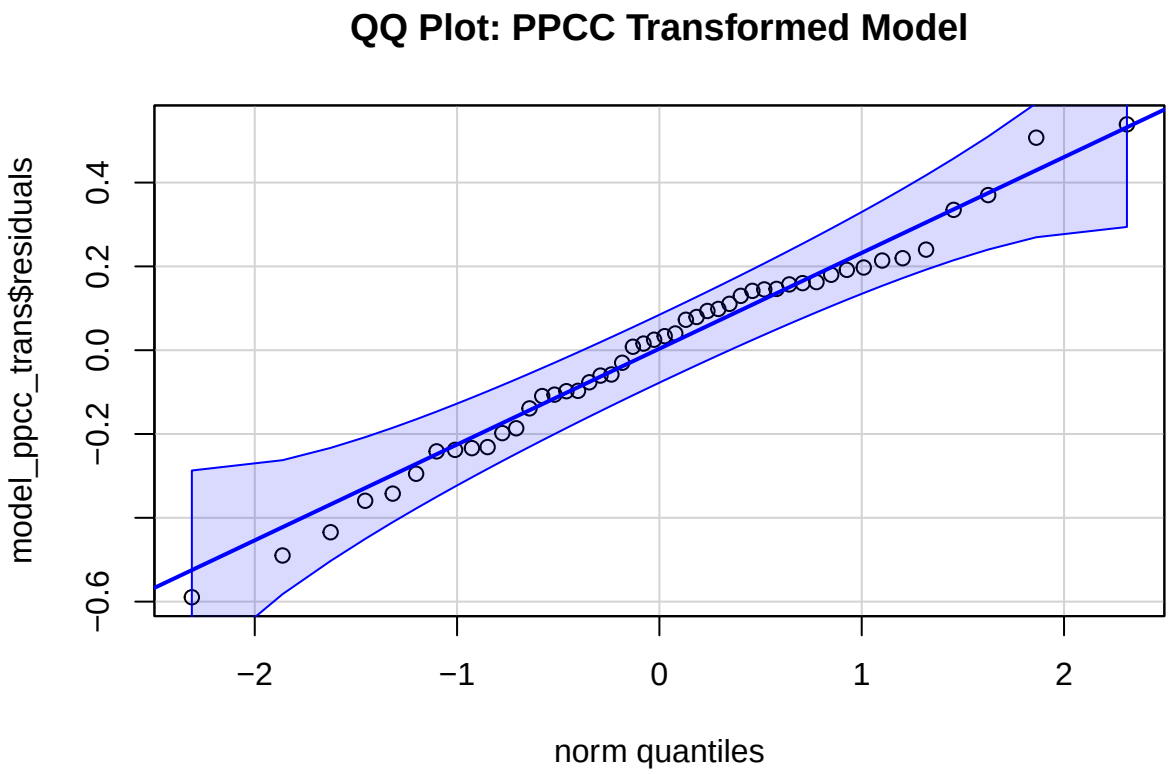
```r
plot(model_log_trans, which = 3, main = "Residual v.s. Fitted: Log Transformed Model")
```




```r
print(leveneTest(model_log_trans))
```

```
##       Df  F value     Pr(>F)
## group 11 1.824302 0.08595076
##       36       NA         NA
```

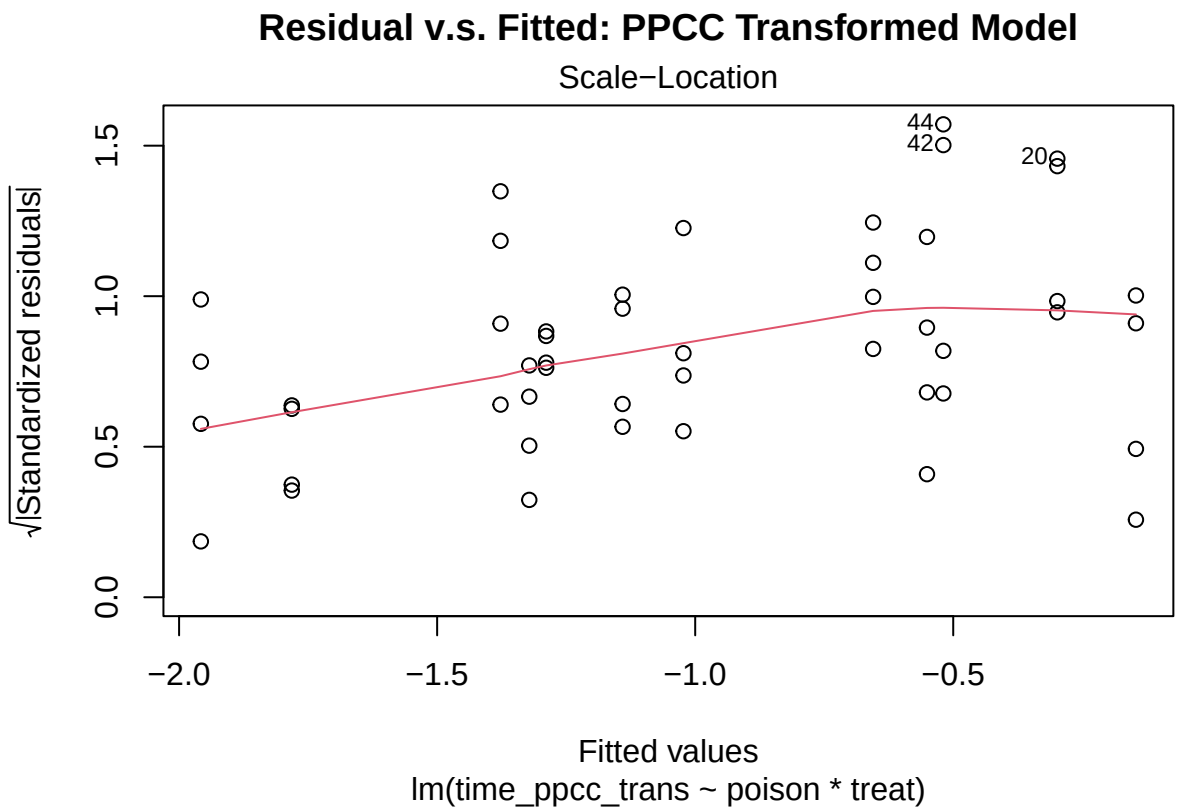
```r
# Test PPCC Transformation
L4 <- boxcox(model_initial, objective.name = "PPCC", optimize = TRUE)$lambda
df$time_ppcc_trans <- (df$time^L4 - 1) / L4
model_ppcc_trans <- lm(time_ppcc_trans ~ poison * treat, data = df)

car::qqPlot(model_ppcc_trans$residuals, id = FALSE, main = "QQ Plot: PPCC Transformed Model")
```




```r
print(shapiro.test(model_ppcc_trans$residuals))
```

```
##
## Results of Hypothesis Test
## --------------------------
##
## Alternative Hypothesis:
##
## Test Name:                      Shapiro-Wilk normality test
##
## Data:                           model_ppcc_trans$residuals
##
## Test Statistic:                 W = 0.9834879
##
## P-value:                        0.7278319
```

```r
plot(model_ppcc_trans, which = 3, main = "Residual v.s. Fitted: PPCC Transformed Model")
```




```r
print(leveneTest(model_ppcc_trans))
```

```
##       Df  F value     Pr(>F)
## group 11 1.983975 0.06018542
##       36       NA         NA
```

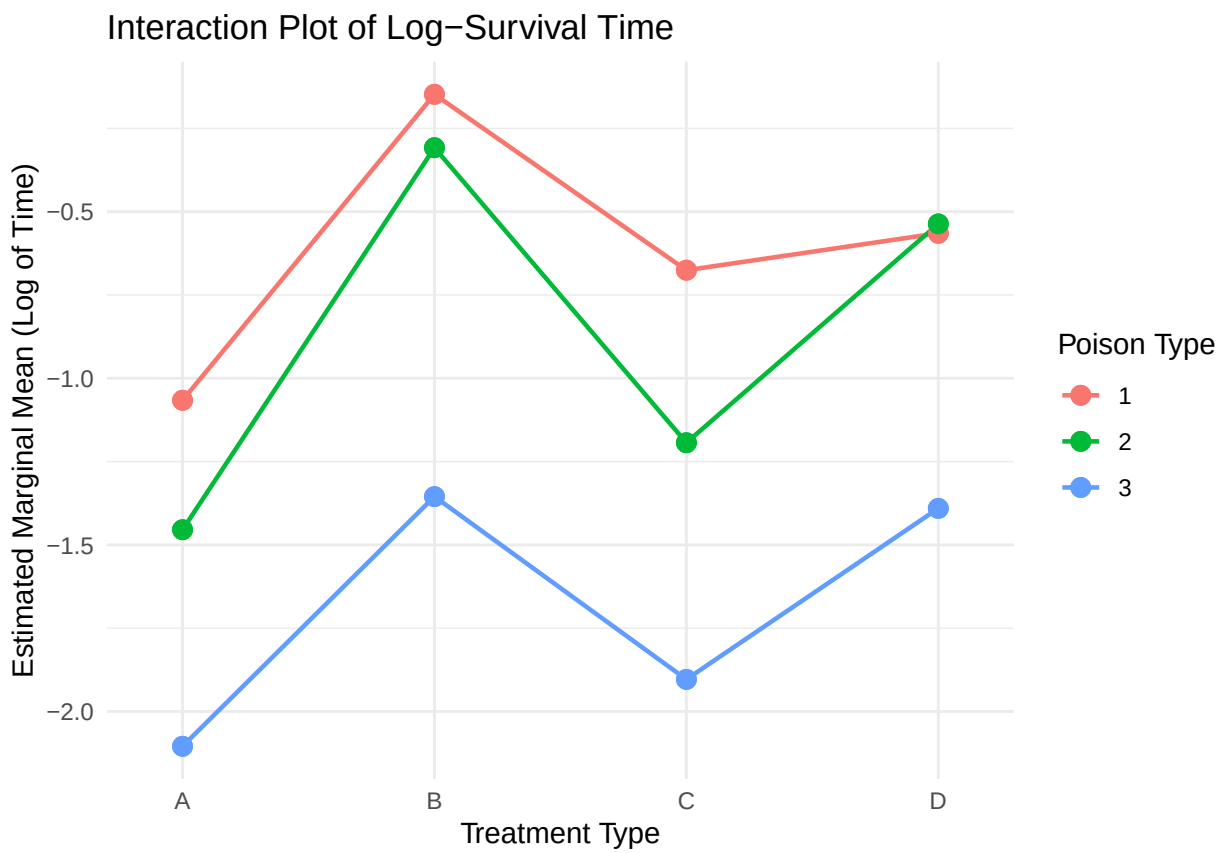
```r
# p-value comparison table
p_shap_initial <- shapiro.test(model_initial$residuals)$p.value
p_shap_ppcc <- shapiro.test(model_ppcc_trans$residuals)$p.value
p_shap_sw <- shapiro.test(model_box_trans$residuals)$p.value
p_shap_ll <- shapiro.test(model_log_trans$residuals)$p.value

p_lev_initial <- leveneTest(model_initial)$"Pr(>F)"[1]
p_lev_ppcc <- leveneTest(model_ppcc_trans)$"Pr(>F)"[1]
p_lev_sw <- leveneTest(model_box_trans)$"Pr(>F)"[1]
p_lev_ll <- leveneTest(model_log_trans)$"Pr(>F)"[1]

diagnostic_summary <- data.frame(
  Transformation = c(
    "None (Initial Model)",
    "PPCC Box-Cox",
    "Shapiro-Wilk Box-Cox",
    "Log-Likelihood"
  ),
  Normality_P = c(p_shap_initial, p_shap_ppcc, p_shap_sw, p_shap_ll),
```

```r
  Variance_P = c(p_lev_initial, p_lev_ppcc, p_lev_sw, p_lev_ll)
)

knitr::kable(
  diagnostic_summary,
  booktabs = TRUE,
  digits = 4,
  caption = "Comparison of Test P-values Across Transformations",
  col.names = c("Transformation Method", "Shapiro-Wilk P-value", "Brown-Forsythe P-value")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 5: Comparison of Test P-values Across Transformations

Transformation Method	Shapiro-Wilk P-value	Brown-Forsythe P-value
None (Initial Model)	0.0016	0.0006
PPCC Box-Cox	0.7278	0.0602
Shapiro-Wilk Box-Cox	0.7496	0.0409
Log-Likelihood	0.6674	0.0860

```r
# select final model based on diagnostics
final_model <- model_log_trans

# 5. Model Selection and Interaction Hypothesis Test

# interaction plot with full model (model_log_trans)
emmip(model_log_trans, poison ~ treat) +
  labs(
    title = "Interaction Plot of Log-Survival Time",
    x = "Treatment Type",
    y = "Estimated Marginal Mean (Log of Time)",
    color = "Poison Type"
  ) +
  theme_minimal()
```




```r
# fit the reduced model (main effects only, no interaction)
model_no_interaction <- lm(time_log_trans ~ poison + treat, data = df)

# ANOVA test to compare the reduced model to the full model
interaction_test <- anova(model_no_interaction, model_log_trans)
knitr::kable(
  interaction_test,
  booktabs = TRUE,
  digits = 4,
  caption = "Hypothesis Test for Interaction (Reduced vs. Full Model)",
  col.names = c("Res.Df", "RSS", "Df", "Sum of Sq", "F value", "Pr(>F)")
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 6: Hypothesis Test for Interaction (Reduced vs. Full Model)

Res.Df	RSS	Df	Sum of Sq	F value	Pr(>F)
42	3.5736	NA	NA	NA	NA
36	3.0751	6	0.4985	0.9726	0.4577

```r
# Partial R square value for interaction: R^2(AB | A+B)
anova_full_model <- anova(model_log_trans)
SS_interaction <- anova_full_model["poison:treat", "Sum Sq"]
SS_error_full <- anova_full_model["Residuals", "Sum Sq"]
```

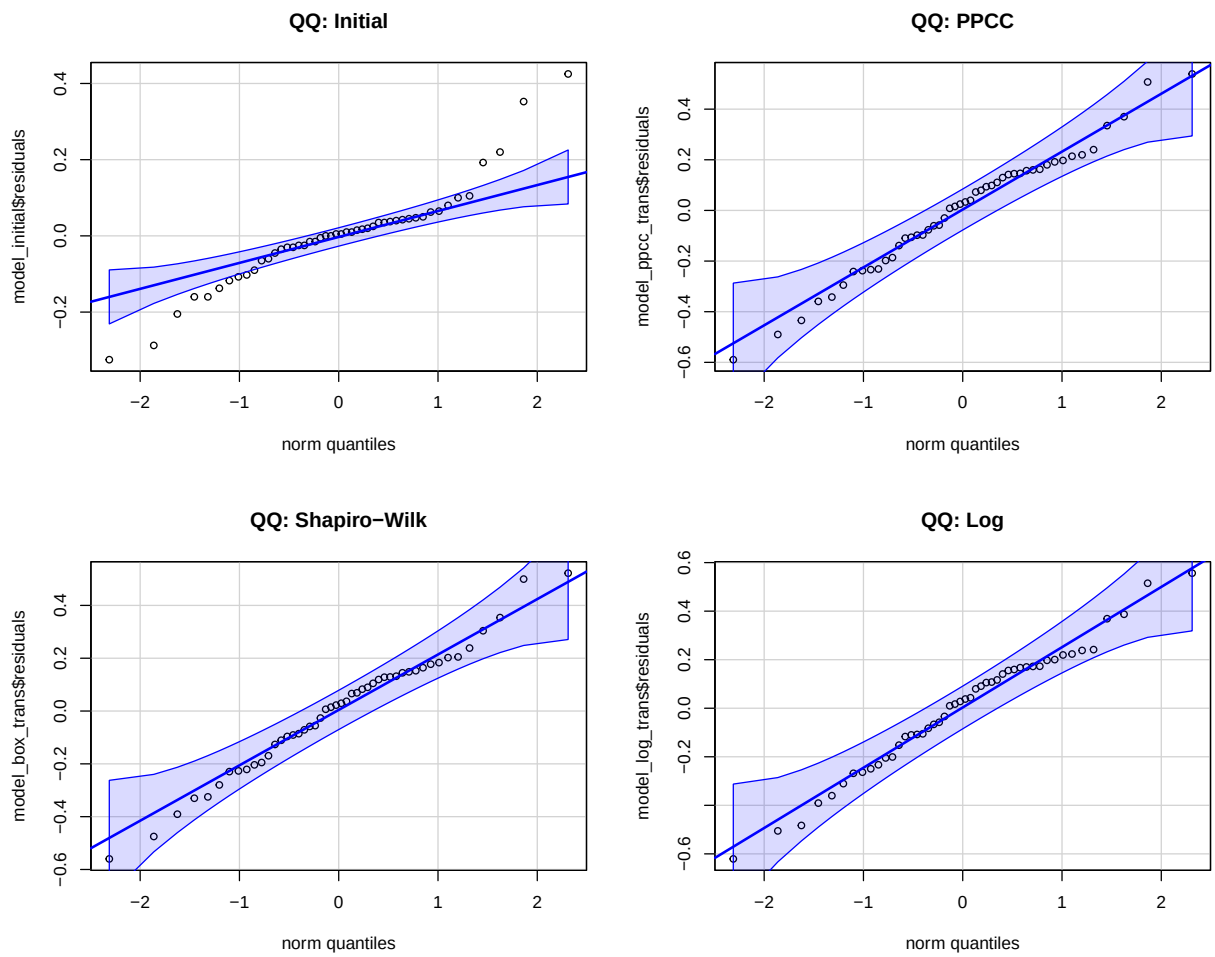
```r
partial_R2_interaction <- SS_interaction / (SS_interaction + SS_error_full)

cat("\n", "Partial R^2 (AB|A+B) for the Interaction Term: ", round(partial_R2_interaction, 4), "\n")
```

```
##
##  Partial R^2 (AB|A+B) for the Interaction Term:  0.1395
```

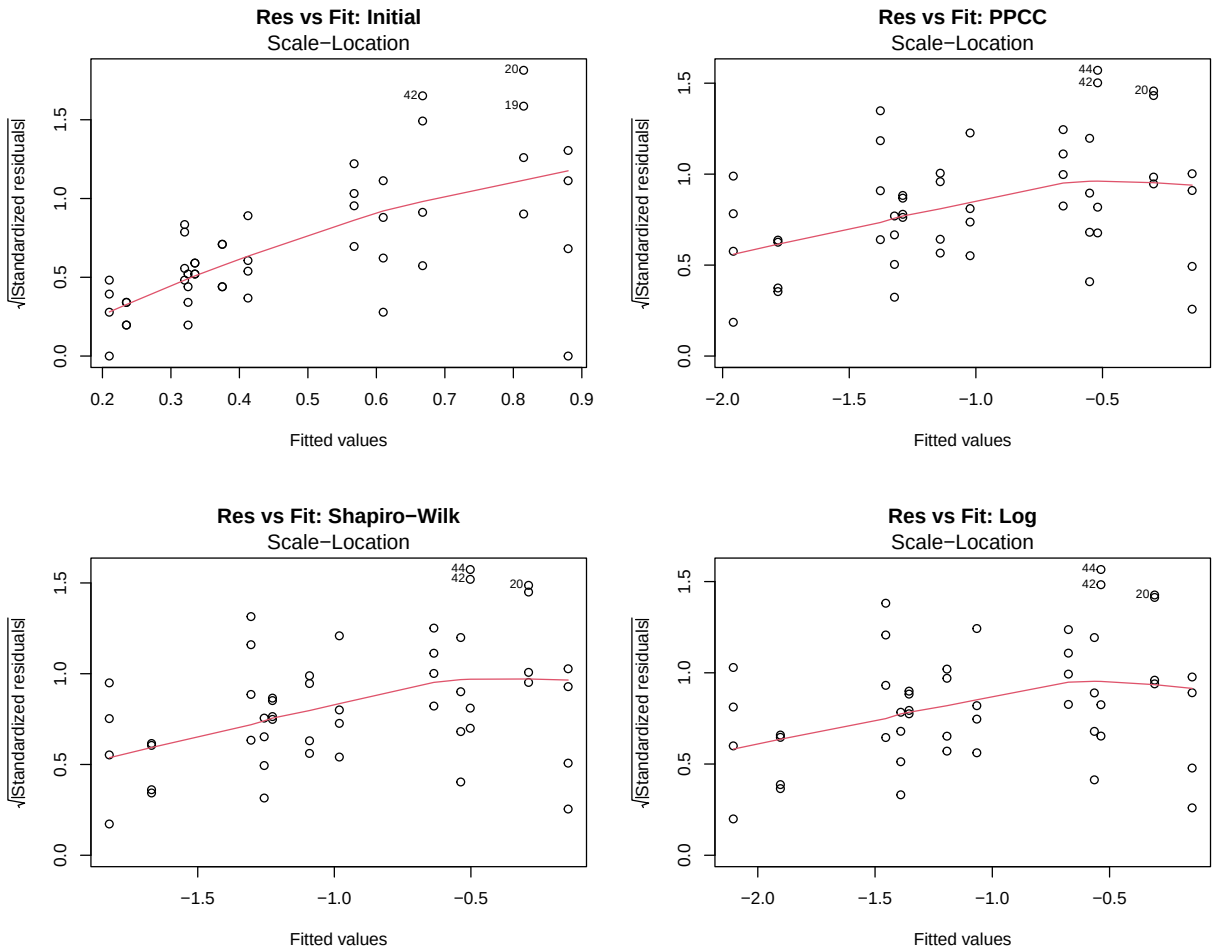
```r
# select final model without interaction as we failed to reject interaction hypothesis
final_additive_model <- aov(time_log_trans ~ poison + treat, data = df)
```

2x2 grid plot graph

```r
# combine qq plots
par(mfrow = c(2, 2))
car::qqPlot(model_initial$residuals, id = FALSE, main = "QQ: Initial")
car::qqPlot(model_ppcc_trans$residuals, id = FALSE, main = "QQ: PPCC")
car::qqPlot(model_box_trans$residuals, id = FALSE, main = "QQ: Shapiro-Wilk")
car::qqPlot(model_log_trans$residuals, id = FALSE, main = "QQ: Log")
```



```r
par(mfrow = c(1, 1))

# combine res vs fit plots
par(mfrow = c(2, 2))
plot(model_initial, which = 3, main = "Res vs Fit: Initial")
plot(model_ppcc_trans, which = 3, main = "Res vs Fit: PPCC")
plot(model_box_trans, which = 3, main = "Res vs Fit: Shapiro-Wilk")
plot(model_log_trans, which = 3, main = "Res vs Fit: Log")
```




```r
par(mfrow = c(1, 1))
```

Analysis

```r
# 1. ANOVA Table
anova_results <- anova(final_additive_model)
knitr::kable(
  anova_results,
  booktabs = TRUE,
  digits = 4,
  caption = "Two-Way ANOVA Table for Final Additive Model"
) %>%
  kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 7: Two-Way ANOVA Table for Final Additive Model

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
poison	2	10.0643	5.0321	59.1416	0
treat	3	6.3821	2.1274	25.0027	0
Residuals	42	3.5736	0.0851	NA	NA

```r
# main effects partial R square R^2(A|B) and R^2(B|A)
SS_poison <- anova_results["poison", "Sum Sq"]
SS_treat <- anova_results["treat", "Sum Sq"]
SS_error_add <- anova_results["Residuals", "Sum Sq"]

partial_R2_poison <- SS_poison / (SS_poison + SS_error_add)
partial_R2_treat <- SS_treat / (SS_treat + SS_error_add)

cat("Partial R^2 (A|B) for Factor A (Poison):", round(partial_R2_poison, 4), "\n")
```

```
## Partial R^2 (A|B) for Factor A (Poison): 0.738
```

```r
cat("Partial R^2 (B|A) for Factor B (Treatment):", round(partial_R2_treat, 4), "\n")
```

```
## Partial R^2 (B|A) for Factor B (Treatment): 0.6411
```

```r
# 2. Confidence Intervals (4 Pairwise, 2 Contrasts)

# --- 4 Pairwise ---
# all pairwise CI
scheffe_pairwise <- ScheffeTest(final_additive_model, conf.level = 0.95)
scheffe_poison_df <- as.data.frame(scheffe_pairwise$poison)
scheffe_treat_df <- as.data.frame(scheffe_pairwise$treat)
scheffe_poison_df$Pair <- rownames(scheffe_poison_df)
scheffe_treat_df$Pair <- rownames(scheffe_treat_df)
combined_scheffe_pairwise <- rbind(scheffe_poison_df, scheffe_treat_df)
combined_scheffe_pairwise <- combined_scheffe_pairwise[, c("Pair", "diff", "lwr.ci", "upr.ci", "pval")]
combined_scheffe_pairwise$pval <- format.pval(combined_scheffe_pairwise$pval, eps = 0.0001, digits = 4)
knitr::kable(
  combined_scheffe_pairwise,
```

```r
  booktabs = TRUE,
  digits = 4,
  row.names = FALSE,
  caption = "All Pairwise Comparisons for Main Effects (Scheffe)",
  col.names = c("Contrast Pair", "Difference", "Lower CI", "Upper CI", "P-Value")
) %>%
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 8: All Pairwise Comparisons for Main Effects (Scheffe)

Contrast Pair	Difference	Lower CI	Upper CI	P-Value
2-1	-0.2595	-0.6196	0.1005	0.2962240
3-1	-1.0748	-1.4348	-0.7147	< 1e-04
3-2	-0.8152	-1.1753	-0.4552	< 1e-04
B-A	0.9380	0.5223	1.3538	< 1e-04
C-A	0.2841	-0.1317	0.6998	0.3553753
D-A	0.7110	0.2952	1.1267	< 1e-04
C-B	-0.6539	-1.0697	-0.2382	0.0002718
D-B	-0.2271	-0.6428	0.1887	0.6070277
D-C	0.4269	0.0111	0.8426	0.0407596

```r
# --- 2 Contrasts (not pairwise) ---
# contrast 1: Poison 1 vs average of Poison 2 and 3 (across all treatments)
# contrast 2: Average of Treat A & C vs Average of Treat B & D (across all poisons)
emm_poison <- emmeans(final_additive_model, ~ poison)
emm_treat <- emmeans(final_additive_model, ~ treat)

contrast_list_poison <- list("Poison 1 vs Poison (2+3)/2" = c(1, -0.5, -0.5))
contrast_list_treat <- list("Treat (A+C)/2 vs Treat (B+D)/2" = c(0.5, -0.5, 0.5, -0.5))

c_poison <- summary(contrast(emm_poison, contrast_list_poison, adjust = "scheffe"), infer = c(TRUE, TRU
c_treat <- summary(contrast(emm_treat, contrast_list_treat, adjust = "scheffe"), infer = c(TRUE, TRUE))

combined_custom <- rbind(as.data.frame(c_poison), as.data.frame(c_treat))

formatted_custom_contrasts <- data.frame(
  Contrast = combined_custom$contrast,
  diff = combined_custom$estimate,
  lwr.ci = combined_custom$lower.CL,
  upr.ci = combined_custom$upper.CL,
  pval = format.pval(combined_custom$p.value, eps = 0.0001, digits = 4)
)

knitr::kable(
  formatted_custom_contrasts,
  booktabs = TRUE,
  digits = 4,
  row.names = FALSE,
  caption = "Custom Non-Pairwise Contrasts (Scheffe)",
  col.names = c("Contrast", "Difference", "Lower CI", "Upper CI", "P-Value")
) %>%
```

```r
kableExtra::kable_styling(latex_options = "hold_position", full_width = FALSE)
```

Table 9: Custom Non-Pairwise Contrasts (Scheffe)

Contrast	Difference	Lower CI	Upper CI	P-Value
Poison 1 vs Poison (2+3)/2	0.6671	0.4869	0.8474	< 1e-04
Treat (A+C)/2 vs Treat (B+D)/2	-0.6824	-0.8524	-0.5125	< 1e-04